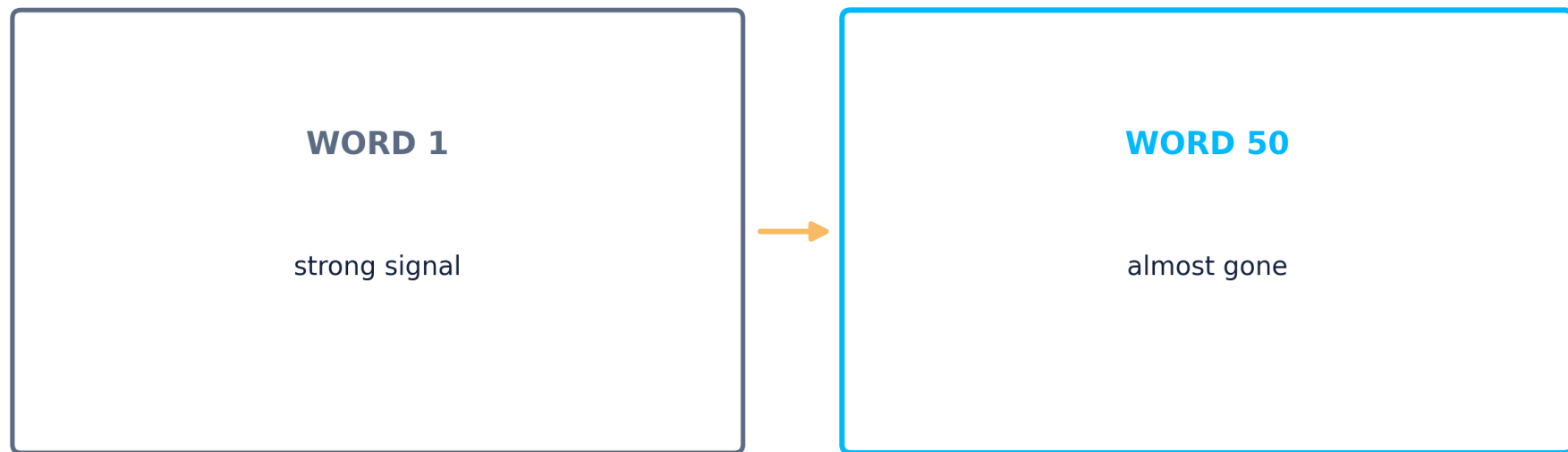


Memory that fades



Early words fade as the sentence grows

An RNN reads a sentence word by word, carrying a **memory** forward. That memory leaks.

- 1 Each word updates a small hidden state passed to the next step.
- 2 By word fifty, the influence of word one has almost vanished.
- 3 That fading is the vanishing gradient — the reason attention was invented.

REMEMBER

RNNs taught us sequence matters. Transformers fixed the forgetting.

